Chapter-5

DIGITAL ARITHMETIC: OPERATIONS AND CIRCUITS

5.1 Introduction

It is essential to know the basics of arithmetic operations such as addition, subtraction, multiplication, and division for a thorough understanding of the operational mechanisms of digital systems such as, digital computers.

5.2 Addition

5.2.1 Binary Addition

In all digital systems, the adder circuit adds only two numbers at a time. When more than two numbers need to be added, the fist two numbers are added and then their result is added to the third number and so on. To understand the addition process, let us consider two single bit numbers. For two 1-bit numbers, there are four combinations for addition as follows.

```
i)          ii)         iii)        iv)
    0           1           0           1     ←— Augend
  + 0         + 0         + 1         + 1     ←— Addend
  ---         ---         ---         ---
   00          01          01          10
  (CS)        (CS)        (CS)        (CS)    ; C=Carry, S=Sum
```

Here C=0, S=0   Here C=0, S=1   Here C=0, S=1   Here C=0, S=1

These four combinations are summarized in the following truth table:

Truth table

Augend, A	Addend, B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Example 1. Add the following binary numbers:

```
i)        ii)        iii)        iv)
   11        101        111        10101
   01        100        101        11111
   --        ---        ---        -----
    ?          ?          ?            ?
```

Solution: i)

1	
1	1
0	1
1 0	1 0
(C_1 S_1)	(C_0 S_0)

Here C_0 is added to the next column digits.
So, result is $C_1S_1S_0$=100.

ii)

1	0	1	← Augend
1	0	0	← Addend
1 0	0	1	
(C_2 S_2)	(C_1 S_1)	(C_0 S_0)	

Here C_0=C_1=0.
So, result is $C_2S_2S_1S_0$=1001.

iii)

1	1	
1	1	1
1	0	1
1 1	1 0	1 0
(C_2 S_2)	(C_1 S_1)	(C_0 S_0)

Here C_0 is added to 2^{nd} col. And C_1 is added to 3^{rd} col.
So, result is $C_2S_2S_1S_0$=1100.

iv)

1	1	1	1	
1	0	1	0	1
1	1	1	1	1
1 1	1 0	1 1	1 0	1 0
(C_4 S_4)	(S_3)	(S_2)	(S_1)	(S_0)

Result is $C_4S_4S_3S_2S_1S_0$=110100.

Example 2. Add the following four binary numbers:

```
 01010
 01101
 10010
 01111
 -----
   ?
```

Solution: Procedure for addition for more than two numbers:

1. For each pair of 1's in a column, add a carry to the next column.
2. If the number of 1's in a column is even, put sum S=0 and for odd number of 1's, put S=1.

	1	1	1			Carries
1	1	1	1	1		
	0	1	0	1	0	
	0	1	1	0	1	
	1	0	0	1	0	
	0	1	1	1	1	
1	1	1	0	0	0	(Result)

Explanation:

i) In column 1, there is only one pair of 1's. So, a carry goes to column 2 and put sum=0.

ii) Now two pair of 1's are in the 2^{nd} column. So, two carries go to 3^{rd} column and put sum=0.

iii) Similarly, two pair of 1's are in the 3^{rd} column. So, two carries goes to 4^{th} column and put again sum=0.

iv) In column 4, there are two pair +1 of 1's. So, two carries goes to column 5 and put sum=1.

v) In column 5, there are one pair +1 of 1's. So, a carry goes to column 6 and put sum=1.

vi) In the 6^{th} column, the sum is 1. Thus, result is 111000.

5.2.2 Octal Addition

Octal addition can be performed using the following steps:

1. Add two octal digits. If the sum is less than or equal to 7, it can be directly expressed as an octal digit.
2. If the sum is greater than 7, subtract 8 from the sum and add a carry to the next column digits.

Example 3. Add the following octal numbers:

i) 53+14=?   ii) 126+237=?   iii) 152+336=?

Solution: i)

5	3	← Augend
1	4	← Addend
6	7	

ii)

1	2	6	← Augend
2	3	7	← Addend
3	5	13	; Since 13>7, subtract 8 from 13 and add a carry to the next column.
	+1	← -8	
3	6	5	

iii)

1	5	2	
3	3	6	
4	8	8	; Since 8>7, subtract 8 and add a carry to the next column.
	+1	← -8	
4	9	0	
+1	← -8		; Since 9>7, subtract 8 and add a carry to the next column.
5	1	0	

5.2.3 Hexadecimal Addition

Hexadecimal addition can be performed using the following steps:

1. Add two hexadecimal digits. If the sum is less than or equal to 15, it can directly be expressed as a hexadecimal digit.
2. If the sum is greater than 15, subtract 16 from the sum and add a carry to the next column digits.

xample 4. Add the following hexadecimal numbers:

i) 67+26=? ii) 2A+88=? iii) 2CF+3B2=?

olution:

i)

6	7
2	6
8	13 (D)

Here 7+6(=13)<15. It is hexadecimal digit D. So, result=8D.

ii)

2	A
8	8
A	18
+1 ←	-16
B	2

; Since A(10)+8=18 >15, subtract 16 from 18 and add a carry to the next column.

iii)

2	C	F
3	B	2
5	23	17
	+1 ←	-16
5	24	1
+1 ←	-16	
6	8	1

; Since F(15)+2=17>15, subtract 16 and add a carry to the next column.

; Since C(12)+B(11)+1=24>15, subtract 16 and add a carry to the next column.

4 BCD Addition

D addition can be performed using the following steps:

1. Add two BCD code groups using binary addition.
2. If the 4-bit sum is less than or equal to 1001 (9 in decimal), it is a valid BCD code. So, no correction is required.
3. If the 4-bit sum is greater than 1001 (9 in decimal), it is an invalid BCD code. So, a correction code 0110 is to be added to get the proper sum. This addition always generates a carry into the next BCD code groups.

uple 5. Add the following BCD numbers:

i)

0010
0110
?

ii)

0001	0111
0010	0101
?	

iii)

1000	1000
0111	1000
?	

Solution:

i)

```
  0010
  0110
 ------
  1000
```

Here sum=1000(<1001). So, result is valid BCD code.

ii)

```
  0001        0111
  0010        0101
 ------------------
  0011        1100
    +1   <-  +0110
 ------------------
  0100     (1)0010
```

Result=0100 0010

; Here 1100>1001. As it is invalid code, the correction code 0110 is added and a carry goes to the next code groups.

iii)

```
        1000       1000
        0111       1000
      ---------------------
        1111      10000
       +0110      +0110
      ---------------------
     (1)0101  <- (1)0110
          +1
   ------------------------
   0001 0110       0110
```

Result=0001 0110 0110

;Here both 1111 and 10000 are greater than 1001. So, correction code 0110 is added to each sum and each carry goes to the next code groups.

5.3 Subtraction

5.3.1 Binary Subtraction

To better understand the subtraction process, let us consider two single-bit numbers: A (Minuend) and B (Subtrahend). For two single-bit numbers, there are four combinations for subtraction as follows:

```
i)          ii)         iii)        iv)
   0           0           1           1    <-- A (Minuend)
 - 0         - 1         - 0         - 1    <-- B (Subtrahend)
 ----        ----        ----        ----
 0 0         1 1         0 1         0 0
(Bo D)      (Bo D)      (Bo D)      (Bo D)      ; D=Difference,
                                                  Bo=Borrow
Here Bo=0,  Here Bo=1,  Here Bo=0,  Here Bo=0,
  D=0         D=1         D=1         D=0
```

It is now clear that when a number B is subtracted from another number A, the result has two parts: a Difference (D) and a Borrow (B). The four combinations for subtraction are summarized in the following truth table:

Truth table

Minuend, A	Subtrahend, B	D	B_o
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Example 6. Perform the following binary subtractions:

i)	ii)	iii)	iv)
11	11	1010	1000
01	10	1001	0111
?	?	?	?

Solution: i)

	1	1
-	0	1
	0 1	0 0
	(B_1 D_1)	(B_0 D_0)

Here Borrows $B_0=B_1=0$.
Result is $D_1D_0=10$.

ii)

	1	1	← Minuend
-	1	0	← Subtrahend
	0 0	0 1	
	(B_1 D_1)	(B_0 D_0)	

Here Borrows $B_0=B_1=0$.
Result is $D_1D_0=01$.

iii)

	1	0	1	0
-	1	0	0+1	1
	0 0	0 0	0 0	1 1
	(B_3 D_3)	(B_2 D_2)	(B_1 D_1)	(B_0 D_0)

Here Borrow $B_0=1$. So it is added to the next subtrahend bit and then subtraction is performed. Result is $D_3D_2D_1D_0=0001$.

iv)

	1	0	0	0
-	0+1	1+1	1+1	1
	0 0	1 0	1 0	1 1
	(B_3 D_3)	(B_2 D_2)	(B_1 D_1)	(B_0 D_0)

Here Borrows $B_0=B_1=B_2=1$. So these are added to their next subtrahend bits and then subtraction is performed. Result is $D_3D_2D_1D_0=0001$.

5.3.2 Octal Subtraction

Octal subtraction can be performed by converting the octal numbers into binary numbers and then using the binary subtraction process.

Example 7. Perform the following octal subtractions:

i) 34-17=? ii) 234-145=?

Solution:

i)

			⇒							
	3	4			0	1	1	1	0	0
-	1	7		-	0	0+1	1+1	1+1	1+1	1
	1	5			0 0	0 0	1 1	1 1	1 0	1 1
					(B_5 D_5)	(B_4 D_4)	(B_3 D_3)	(B_2 D_2)	(B_1 D_1)	(B_0 D_0)

Result is $D_5D_4D_3D_2D_1D_0=001\ 101$ (15 in Octal).

i)	ii)	iii)	iv)
11	11	1010	1000
01	10	1001	0111
?	?	?	?

Solution:

i)

	1	1	
-	0	1	
	0 1	0 0	
	(B_1 D_1)	(B_0 D_0)	

Here Borrows B_0=B_1=0.
Result is D_1D_0=10.

ii)

	1	1	← Minuend
-	1	0	← Subtrahend
	0 0	0 1	
	(B_1 D_1)	(B_0 D_0)	

Here Borrows B_0=B_1=0.
Result is D_1D_0=01.

iii)

	1	0	1	0
-	1	0	0+1	1
	0 0	0 0	0 0	1 1
	(B_3 D_3)	(B_2 D_2)	(B_1 D_1)	(B_0 D_0)

Here Borrow B_0=1. So it is added to the next subtrahend bit and then subtraction is performed. Result is $D_3D_2D_1D_0$=0001.

iv)

	1	0	0	0
-	0+1	1+1	1+1	1
	0 0	1 0	1 0	1 1
	(B_3 D_3)	(B_2 D_2)	(B_1 D_1)	(B_0 D_0)

Here Borrows B_0=B_1=B_2=1. So these are added to their next subtrahend bits and then subtraction is performed. Result is $D_3D_2D_1D_0$=0001.

5.3.2 Octal Subtraction

Octal subtraction can be performed by converting the octal numbers into binary numbers and then using the binary subtraction process.

Example 7. Perform the following octal subtractions:

i) 34-17=? ii) 234-145=?

Solution:

i)

3 4	⇒		0	1	1	1	0	0
- 1 7		-	0	0+1	1+1	1+1	1+1	1
1 5			0 0	0 0	1 1	1 1	1 0	1 1
			(B_5 D_5)	(B_4 D_4)	(B_3 D_3)	(B_2 D_2)	(B_1 D_1)	(B_0 D_0)

Result is $D_5D_4D_3D_2D_1D_0$=001 101 (15 in Octal).